

# The diagonal expectation is locally but not quantum factorisable

FA/Banach run `fa_banach_001`, lane 1

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## Source metadata and status

The source is A. Chatzinikolaou, I. G. Todorov, and L. Turowska, *Absolutely dilatable bimodule maps*, arXiv:2411.08086 (2024), published in *International Mathematics Research Notices* (2025), article rna160 [1]. The question occurs across pages 25–26 of the arXiv PDF, immediately after equation (28). A local copy is stored as `source_paper.pdf`.

**Claimed status: counterexample, likely valid.** The inclusion in equation (28) fails for an infinite-dimensional separable Hilbert space, already for the discrete maximal abelian von Neumann algebra.

## The source question

For a von Neumann algebra  $\mathcal{D} \subseteq \mathcal{B}(H)$ , the source writes  $\mathfrak{D}_{\text{loc}, \mathcal{D}}(H)$  for the absolutely dilatable  $\mathcal{D}'$ -bimodule maps admitting an abelian ancilla, and  $\mathfrak{D}_{\text{q}, \mathcal{D}}(H)$  for those admitting a finite-dimensional ancilla. It proves the inclusion in finite dimensions and asks whether

$$\mathfrak{D}_{\text{loc}, \mathcal{D}}(H) \subseteq \mathfrak{D}_{\text{q}, \mathcal{D}}(H) \tag{1}$$

also holds when  $H$  is infinite-dimensional.

From the previous theorem, we have  $\mathfrak{D}_{\text{loc}, \mathcal{D}} = \overline{\text{conv}}(\text{Aut}_{\mathcal{D}'}(\mathcal{B}(H)))$ . The elements of  $\text{Aut}_{\mathcal{D}'}(\mathcal{B}(H))$  belong to  $\mathfrak{D}_{\text{q}, \mathcal{D}}(H)$  (with one-dimensional ancilla); thus, by Proposition 3.1

$$\mathfrak{D}_{\text{loc}, \mathcal{D}}(H) \subseteq \overline{\mathfrak{D}_{\text{q}, \mathcal{D}}(H)} \text{ and } \mathfrak{D}_{\text{q}, \mathcal{D}}(H) \subseteq \mathfrak{D}_{\text{qc}, \mathcal{D}}(H).$$

If the Hilbert space  $H$  is finite dimensional, then the convex hull of  $\text{Aut}_{\mathcal{D}'}(\mathcal{B}(H))$  is closed and hence

$$(28) \quad \mathfrak{D}_{\text{loc}, \mathcal{D}}(H) \subseteq \mathfrak{D}_{\text{q}, \mathcal{D}}(H).$$

## Proof intuition

Use the diagonal conditional expectation on  $\mathcal{B}(\ell^2)$ . It is a continuous random mixture of diagonal unitary conjugations: average conjugation by  $\text{diag}(1, z, z^2, \dots)$  over the circle. This gives an abelian ancilla and hence local factorisability.

The same map has Schur symbol  $[\delta_{ij}]$ . A finite-dimensional quantum ancilla would represent this symbol as  $\delta_{ij} = \tau(v_i^* v_j)$  for unitaries  $v_i$  in a single finite-dimensional tracial algebra. Thus the  $v_i$  would be infinitely many orthonormal vectors in its finite-dimensional  $L^2$ -space. The contradiction is the finite Gram-rank obstruction in its simplest form.

## Counterexample theorem

**Theorem 1.** *Let  $H = \ell^2(\mathbb{N}_0)$  and let  $\mathcal{D} = \ell^\infty(\mathbb{N}_0) \subseteq \mathcal{B}(H)$  be the diagonal masa. The normal diagonal conditional expectation*

$$\Phi : \mathcal{B}(H) \longrightarrow \mathcal{D}, \quad \Phi([x_{ij}]) = \text{diag}(x_{00}, x_{11}, \dots),$$

*satisfies*

$$\Phi \in \mathfrak{D}_{\text{loc}, \mathcal{D}}(H) \setminus \mathfrak{D}_{\text{q}, \mathcal{D}}(H).$$

*Consequently, the infinite-dimensional inclusion (1) is false.*

## Proof

Let  $(e_n)_{n \in \mathbb{N}_0}$  be the standard orthonormal basis of  $H$ , let  $p_n = e_n e_n^*$ , and let  $z : \mathbb{T} \rightarrow \mathbb{C}$  be the coordinate function. Equip  $L^\infty(\mathbb{T})$  with normalized Haar integration  $\tau(f) = \int_{\mathbb{T}} f$ . The strong diagonal sum

$$U = \sum_{n=0}^{\infty} p_n \otimes z^n \in \mathcal{D} \overline{\otimes} L^\infty(\mathbb{T})$$

is unitary. Define

$$\Psi(x) = (\text{id} \otimes \tau)(U^*(x \otimes 1)U), \quad x \in \mathcal{B}(H).$$

For the matrix unit  $e_{ij} = e_i e_j^*$ ,

$$U^*(e_{ij} \otimes 1)U = e_{ij} \otimes z^{j-i}, \quad \Psi(e_{ij}) = \left( \int_{\mathbb{T}} z^{j-i} \right) e_{ij} = \delta_{ij} e_{ij}.$$

Both  $\Psi$  and the diagonal expectation  $\Phi$  are normal, so equality on the matrix units gives  $\Psi = \Phi$ . The displayed representation is an exact factorisation through the abelian tracial ancilla  $L^\infty(\mathbb{T})$ . Hence  $\Phi \in \mathfrak{D}_{\text{loc}, \mathcal{D}}(H)$ .

Suppose, toward a contradiction, that  $\Phi \in \mathfrak{D}_{\text{q}, \mathcal{D}}(H)$ . The discrete masa is the multiplication algebra of the standard measure space  $\mathbb{N}_0$ . The type-preserving exactness assertion in the proof of Corollary 3.5 of the source states that, for such a masa,

$$\mathfrak{D}_{\text{t}, \mathcal{D}}(H) = \mathfrak{D}_{\text{t}, \mathcal{D}}^{\text{ex}}(H), \quad \text{t} \in \{\text{loc}, \text{q}, \text{qa}, \text{qc}\}.$$

Therefore there are a finite-dimensional tracial von Neumann algebra  $(\mathcal{M}, \tau_{\mathcal{M}})$  and a unitary  $V \in \mathcal{D} \overline{\otimes} \mathcal{M}$  such that

$$\Phi(x) = (\text{id} \otimes \tau_{\mathcal{M}})(V^*(x \otimes 1)V).$$

Identifying  $\mathcal{D} \overline{\otimes} \mathcal{M}$  with  $\ell^\infty(\mathbb{N}_0; \mathcal{M})$ , write

$$V = \sum_{n=0}^{\infty} p_n \otimes v_n, \quad v_n \in \mathcal{U}(\mathcal{M}).$$

Evaluation on  $e_{ij}$  now gives

$$\delta_{ij} e_{ij} = \Phi(e_{ij}) = \tau_{\mathcal{M}}(v_i^* v_j) e_{ij}, \quad i, j \in \mathbb{N}_0.$$

Thus  $(v_n)_{n \in \mathbb{N}_0}$  is an infinite orthonormal family in the Hilbert space  $L^2(\mathcal{M}, \tau_{\mathcal{M}})$ . This is impossible because  $\mathcal{M}$ , and hence its  $L^2$ -space, is finite-dimensional. Therefore  $\Phi \notin \mathfrak{D}_{\mathfrak{q}, \mathcal{D}}(H)$ , completing the proof.  $\square$

## Verification notes

The proof has no numerical or computer-assisted component. Its four checks are exact:

- the diagonal field  $(z^n)_{n \geq 0}$  defines a unitary in the stated von Neumann tensor product;
- Haar orthogonality gives the Schur symbol  $\delta_{ij}$ ;
- the source's proof of Corollary 3.5 explicitly includes  $t = q$  in the equality between factorisable and exactly factorisable maps for a standard masa;
- the unitary coefficients of a finite-dimensional exact ancilla live in a finite-dimensional tracial  $L^2$ -space.

## Scope, novelty check, and review recommendation

The example disproves the universal infinite-dimensional inclusion asked in equation (28), already for  $H = \ell^2$  and an atomic masa. It does not classify which other pairs  $(H, \mathcal{D})$  may still satisfy the inclusion.

The run indexes were searched for arXiv:2411.08086, the paper title, the exact inclusion, local and quantum factorisability, Schur multipliers, and diagonal conditional expectations. Bounded web searches on July 21, 2026 used the exact question and notation, the source title, and diagonal-expectation and finite-ancilla variants. The arXiv paper and its 2025 IMRN publication were found; the published version still states the question as unknown. No later answer, correction, or matching counterexample was found. Novelty confidence is moderate because the obstruction is elementary and could be known informally.

**Human review recommendation.** Review as a likely valid full counterexample. The main point to verify against the source's definitions is the type-preserving exactness sentence in the proof of Corollary 3.5. Once that sentence is accepted in its stated scope, the orthogonality contradiction is immediate.

## References

- [1] A. Chatzinikolaou, I. G. Todorov, and L. Turowska, *Absolutely dilatable bimodule maps*, arXiv:2411.08086 (2024); International Mathematics Research Notices (2025), rnafl60, doi:10.1093/imrn/rnafl60.